FirmGuard security report

Scan summary

Field	Value
Device	Sample router
Version / revision	1.0 / 1
Scan / created (UTC)	1 / 2026-11-02T00:00:00+00:00
Input / bytes	firmware_v1.zip / 926
Scanner / advisory rules	firmguard-1.0-rules-1 / 2026-11-02-curated-1
Extraction / coverage	native / supported checks completed
Files / text / binary	6 / 5 / 1
Findings	13

Integrity

SHA-256: ce9001b4c291a12014b28004ac36c7a7f8babf936e22ca21c89fff0ed1e9961c

Reference status: not_compared

Expected digest: not supplied

Scope and limitations

Static pattern and metadata analysis only. No findings does not prove security. Hash equality does not establish authenticity. Version matches require vendor verification.

Software inventory

Component	Version	File / evidence	Check
OpenSSL	1.0.1f	usr/lib/version.txt / banner	potential_match
BusyBox	1.30.1	usr/lib/version.txt / banner	not_covered
dnsmasq	2.77	usr/lib/version.txt / banner	potential_match

ELF security properties

usr/bin/sample.elf

class: 64; endian: little; machine: EM_X86_64; type: ET_EXEC; nx: disabled; pie: disabled; relro: none; stack_canary: unknown; fortify: unknown; status: analysed

Findings and actions

1. [High] Hardcoded Wi-Fi password

Rule: CRED-001 | File: etc/device.conf | Line: 2
A literal credential assignment was detected. Its value is hidden.
Action: Remove embedded secrets, rotate exposed credentials and use protected provisioning.
Assignment: wifi_password=********

2. [High] Hardcoded password

Rule: CRED-001 | File: etc/device.conf | Line: 3
A literal credential assignment was detected. Its value is hidden.
Action: Remove embedded secrets, rotate exposed credentials and use protected provisioning.
Assignment: password=********

3. [High] Telnet enabled

Rule: CONF-TELNET | File: etc/device.conf | Line: 4
An explicit setting matches this rule. Includes, defaults and runtime overrides are not resolved.
Action: Disable Telnet and use SSH.

4. [High] Certificate verification disabled

Rule: CONF-TLS | File: etc/device.conf | Line: 5
An explicit setting matches this rule. Includes, defaults and runtime overrides are not resolved.
Action: Enable certificate and hostname verification.

5. [Medium] Debug mode enabled

Rule: CONF-DEBUG | File: etc/device.conf | Line: 6
An explicit setting matches this rule. Includes, defaults and runtime overrides are not resolved.
Action: Disable debug mode in released firmware.

6. [High] SSH root login enabled

Rule: CONF-SSH-ROOT | File: etc/ssh/sshd_config | Line: 1
An explicit setting matches this rule. Includes, defaults and runtime overrides are not resolved.
Action: Disable direct root login.

7. [Medium] SSH password authentication enabled

Rule: CONF-SSH-PASSWORD | File: etc/ssh/sshd_config | Line: 2
An explicit setting matches this rule. Includes, defaults and runtime overrides are not resolved.
Action: Prefer provisioned SSH keys where feasible.

8. [High] Executable stack requested

Rule: ELF-NX | File: usr/bin/sample.elf | Line: -
PT_GNU_STACK has the execute flag.
Action: Link with a non-executable stack where supported.

9. [Medium] Executable without PIE

Rule: ELF-PIE | File: usr/bin/sample.elf | Line: -
The executable uses ET_EXEC rather than PIE.
Action: Build supported executables with -fPIE and -pie.

10. [Medium] Full RELRO not detected

Rule: ELF-RELRO | File: usr/bin/sample.elf | Line: -
GNU_RELRO and immediate binding were not both present.
Action: Link supported programs with -Wl,-z,relro,-z,now.

11. [High] Account with empty password field

Rule: CONF-ACCOUNT-EMPTY | File: etc/passwd | Line: 1
An account record has an empty password field. Account names are withheld.
Action: Lock unused accounts and require authentication.

12. [High] Potential CVE-2014-0160 version match

Rule: SW-CVE-2014-0160 | File: usr/lib/version.txt | Line: -
Detected OpenSSL 1.0.1f; the upstream range is 1.0.1 to before 1.0.1g. Vendor backports and build options require verification.
Action: Check the vendor advisory and update to a supported, patched build.
Reference: https://www.cve.org/CVERecord?id=CVE-2014-0160

13. [High] Potential CVE-2017-14491 version match

Rule: SW-CVE-2017-14491 | File: usr/lib/version.txt | Line: -
Detected dnsmasq 2.77; the upstream range is 2.0 to before 2.78. Vendor backports and build options require verification.
Action: Check the vendor advisory and update to a supported, patched build.
Reference: https://www.cve.org/CVERecord?id=CVE-2017-14491

File manifest

Path	Bytes / mode	SHA-256
etc/device.conf	130 / 0o644	b18c247c4dee5f1244c90a18a93fe79974a3a35d6e20575c098eefb106348ce5
etc/ssh/sshd_config	47 / 0o644	c6f304501dd051cc53700f7047111f474231dbb6070a1c049f629644753a7230
usr/lib/version.txt	44 / 0o644	469fd97e24c7691396f1cb5b9c5eb9f30537dd6a114e325eb83824f493aabefe
usr/bin/sample.elf	120 / 0o644	fe82db5160fe8e5b6d41b1a56947fd6e12fd5de79672dcc7113fd37a4402db86
etc/passwd	29 / 0o644	45c3e45bafed64c96cb286143358416ffa67c03b656eba99057ec9aa0299907f
etc/old.conf	16 / 0o644	a438ddcb2158f3ffb1bb89d781014ff99c48da885cf966b0c3f002aa40366aaa